Danna Xue

Postdoctoral Researcher

Computer Vision Center (CVC), Barcelona, Spain

+34 625589146 | dxue@cvc.uab.es | dxue321.github.io | dxue321 | dxue100294 | Danna Xue

Overview

Currently, I am a postdoctoral researcher in computer vision, especially focusing on image understanding and low-level vision tasks. I have developed novel approaches to solve different problems ranging from semantic segmentation, to image restoration and color-related problems such as color harmonization or re-coloring. I have published 6 papers in top-ranked journals and 3 top-ranked conferences. I have research experience in cutting-edge research labs in Spain, China, and Canada, working in high-profile research projects.

My research interests mainly lie on: Image understanding (Image segmentation); Low-level vision (Image restoration); Color (Gamut mapping, Color editing); Domain Adaptation.

Experience

Universitat Autònoma de Barcelona, Computer Vision Center *Barcelona, Spain*

Postdoctoral Researcher in Computer Vision *Jul 2024 - present*

- High-fidelity color processing in smartphone ISP, e.g. gamut mapping, tone mapping
- Content-aware and consistent video editing

Education

Computer Vision Center, Universitat Autònoma de Barcelona | Northwestern Polytechnical University *Barcelona, Spain | Xi'an, China*

Ph.D. in Computer Science (GPA: 89.25/100) *Mar 2019 - Jul 2024*

- Under joint supervision of ***Prof. Yanning Zhang***, ***Assoc. Prof. Javier Vazquez-Corral***, and ***Dr. Luis Herranz***
- Passed with Cum Laude Mention and supported by the CSC scholarship

Northwestern Polytechnical University *Xi'an, China*

M.Phil. in Guidance, Navigation and Control (GPA: 83.16/100) *Sep 2015 - Apr 2018*

- Specialised in control theory and image processing

Northwestern Polytechnical University *Xi'an, China*

B.Eng. in Detection, Guidance, and Control Technology (GPA: 87.21/100) *Sep 2011 - Jul 2015*

- Specialised in Control theory and Computer Science
- Exchange in Universidad Politécnica de Madrid for 6 months supported by the CSC Scholarship

Research Experience

Computer Vision Center *Barcelona, Spain*

Postdoctoral Researcher, Collaborators: Assoc. Prof. Javier Vazquez-Corral, Dr. Shaolin Su, Mr. David Serrano-Lozano, Ms. Maria Pilligua Costa *Jul 2024 - present*

- **High-fidelity color processing in smartphone ISP**
- Develop an efficient framework for different stages of the color processing pipeline in smartphone ISP.
- **Consistent video editing**
- Accelerating the neural video decomposition model with hypernetwork. (**CVPR'25**)

York University *Toronto, Canada*

Research stay with Prof. Michael S. Brown *Feb 2023 - Apr 2023*

- **Palette-based image color manipulation** (**PG'23**, **IEEE-SPL**)
- Develop an intuitive, user-friendly, and content-aware framework for single and multiple image color editing.
- Leverage color-naming for perceptual multi-image color consistency and single image color harmonization.

Computer Vision Center *Barcelona, Spain*

Ph.D student with Assoc. Prof. Javier Vazquez-Corral and Dr. Luis Herranz *May 2021 - Jul 2024*

- **Perception-distortion tradeoff for multi-image restoration** (**ICASSP'23**)
- Extend the perception-distortion tradeoff theory by introducing multiple-frame information.
- Propose the area of the unattainable region as a new metric for PD tradeoff evaluation.
- Analyse the performance of burst restoration under both aligned bursts and misaligned bursts situations.
- **Slimmable semantic segmentation** (**ACM MM'22**)
- Propose an effective slimmable semantic segmentation method that can be executed at different capacities during inference.
- Design parametrized channel slimming by stepwise downward knowledge distillation for model training.

Multimedia Lab, Northwestern Polytechnical University *Xi'an, China*
Ph.D student, Collaborators: Dr. Rui Li, and Dr. Pei Wang *Mar 2019 - May 2021*

- **Semantic aware image processing and understanding.**
- Propose a self-supervised monocular depth estimation method with semantic guidance. (**PR**)
- Propose a severe blur removal method using semantic priors. (**CVIU**)

Publications

Journal (3 Q1, 2 Q2)

Palette-based Color Harmonization via Color Naming [Code] [Paper]
Danna Xue, Javier Vazquez-Corral, Luis Herranz, Yanning Zhang, Michael S Brown
IEEE Signal Processing Letters (2024). IEEE, 2024

Take a prior from other tasks for severe blur removal [arXiv]
Pei Wang, Yu Zhu, Danna Xue, Qingsen Yan, Jinqiu Sun, Sung-eui Yoon, Yanning Zhang
Computer Vision and Image Understanding (2024) p. 104027. Elsevier, 2024

Integrating high-level features for consistent palette-based multi-image recoloring [Paper]
Danna Xue, J Vazquez Corral, Luis Herranz, Yanning Zhang, Michael S Brown
Computer Graphics Forum, 2023

Learning depth via leveraging semantics: Self-supervised monocular depth estimation with both implicit and explicit semantic guidance [Paper]
Rui Li, Danna Xue, Shaolin Su, Xiantuo He, Qing Mao, Yu Zhu, Jinqiu Sun, Yanning Zhang
Pattern Recognition 137 (2023) p. 109297. Elsevier, 2023

Self-supervised monocular depth estimation with frequency-based recurrent refinement [Paper]
Rui Li, Danna Xue, Yu Zhu, Hao Wu, Jinqiu Sun, Yanning Zhang
IEEE Transactions on Multimedia 25 (2022) pp. 5626–5637. IEEE, 2022

Conference (2 Class-1, 1 Class-2 —GII-GRIN-SCIE ranking—)

HyperNVD: Accelerating Neural Video Decomposition via Hypernetworks (CCF-A) [Project] [arXiv]
Maria Pilligua*, Danna Xue*, Javier Vazquez-Corral (*Equal Contribution)
International Conference on Computer Vision and Pattern Recognition (CVPR), 2025

Burst perception-distortion tradeoff: analysis and evaluation (CCF-B) [Paper]
Danna Xue, Luis Herranz, Javier Vazquez Corral, Yanning Zhang
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023

Slimseg: Slimmable semantic segmentation with boundary supervision (CCF-A) [arXiv]
Danna Xue, Fei Yang, Pei Wang, Luis Herranz, Jinqiu Sun, Yu Zhu, Yanning Zhang
ACM International Conference on Multimedia (ACMMM), 2022

Preprint

Rethinking Image Evaluation in Super-Resolution [Project] [arXiv]
Shaolin Su, Josep M Rocafort, Danna Xue, David Serrano-Lozano, Lei Sun, Javier Vazquez-Corral
arXiv preprint arXiv:2503.13074 (2025). 2025

Synth-to-Real Unsupervised Domain Adaptation for Instance Segmentation [arXiv]
Yachan Guo, Yi Xiao, Danna Xue, Jose Luis Gomez Zurita, Antonio M López
arXiv preprint arXiv:2405.09682 (2024). 2024

Academic Services

Conference Review CVPR (2025), ECCV (2024), WACV (2023), ACM MM (2023, 2024).
Journal Review Pattern Recognition; IEEE Transactions on Geoscience and Remote Sensing.
Program Committee Session Chair, CVCR&D 2024; Poster Session Chair, ACMCV 2024.

Skills

Programming Python (PyTorch, Tensorflow, NumPy, Scikit-learn, Pandas, Colour, Matplotlib, etc.), MATLAB.
Miscellaneous Linux, Shell, LaTeX(Overleaf/R Markdown), Microsoft Office, Git.
Language **Mandarin** (Native), **English** (Professional, C1 level), **Spanish** (Elementary).